Sequestration of Imaging Studies in MIDRC: Using load factor to minimize algorithm performance overestimation and image reuse

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Conflict of Interest Statement

Dylan Tang: Nothing to Disclose
Heather Whitney: Nothing to disclose
Maryellen L. Giger: Stockholder and receives royalties, Hologic, Inc.; Equity holder and co-founder, Quantitative Insights, Inc. (now consultant to Qlarity Imaging); Shareholder, QView Medical, Inc.; Royalties, General Electric Company, MEDIAN Technologies, Riverain Technologies, LLC, Mitsubishi, and Toshiba.
Kyle J. Myers: Puente Solutions LLC, works as an Independent Consultant to innovators seeking to gain US market authorization for medical imaging devices.

Introduction

MIDRC
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The Medical Imaging and Data Resource Center (MIDRC)

Medical Imaging Community Response to the COVID-19 Pandemic

MIDRC is

- A curated image and data commons
- A means by which researchers can address topics no single image archive could yield independently
- A partnership of the AAPM, ACR and RSNA, supported by NIBIB and hosted at University of Chicago

There are two scientific components of MIDRC:

1. *Open Discovery Data Commons*

   Creation, testing, quality assurance, and data connectivity

2. *Machine Intelligence Computational Capabilities*

   Clinically relevant algorithms and software tools

Established August 2020
Supported by NIBIB

A collaboration of 24+ institutions & 100+ investigators from academia, community practices, FDA, and others

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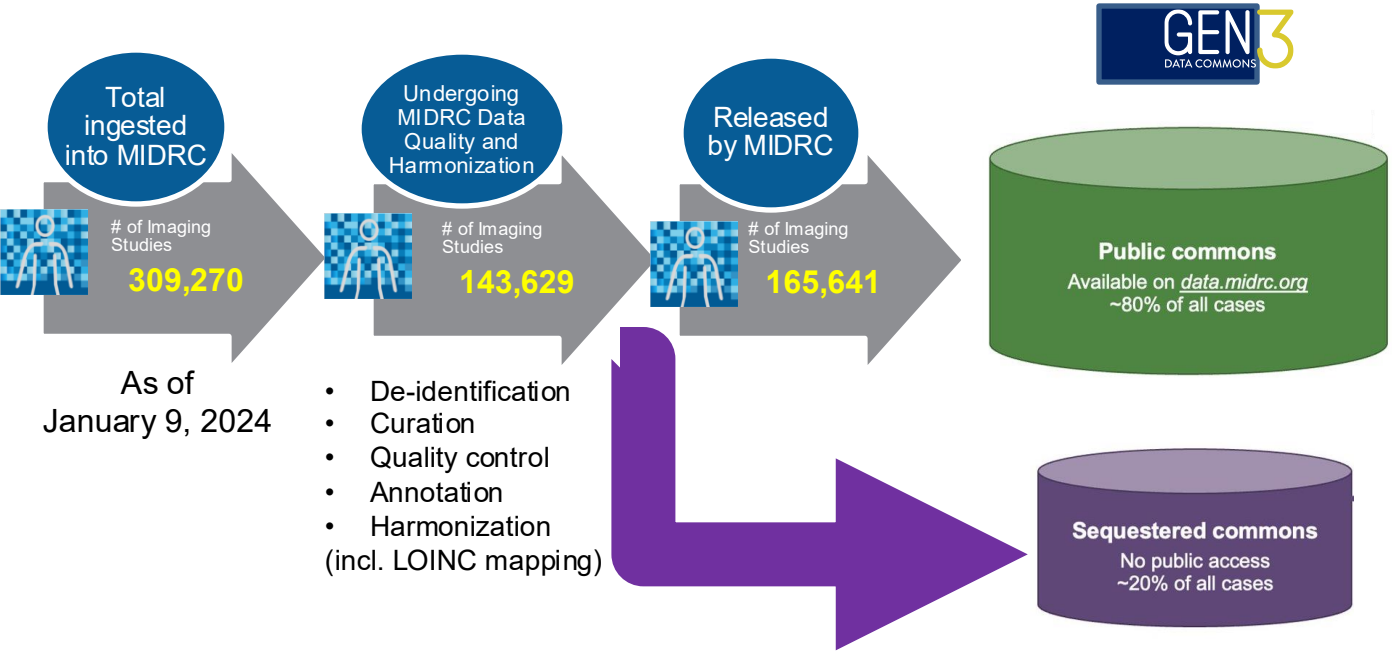
MIDRC facilitates translation of AI algorithms via the sequestered data commons



As of January 9, 2024

- De-identification
- Curation
- Quality control
- Annotation
- Harmonization

(incl. LOINC mapping)

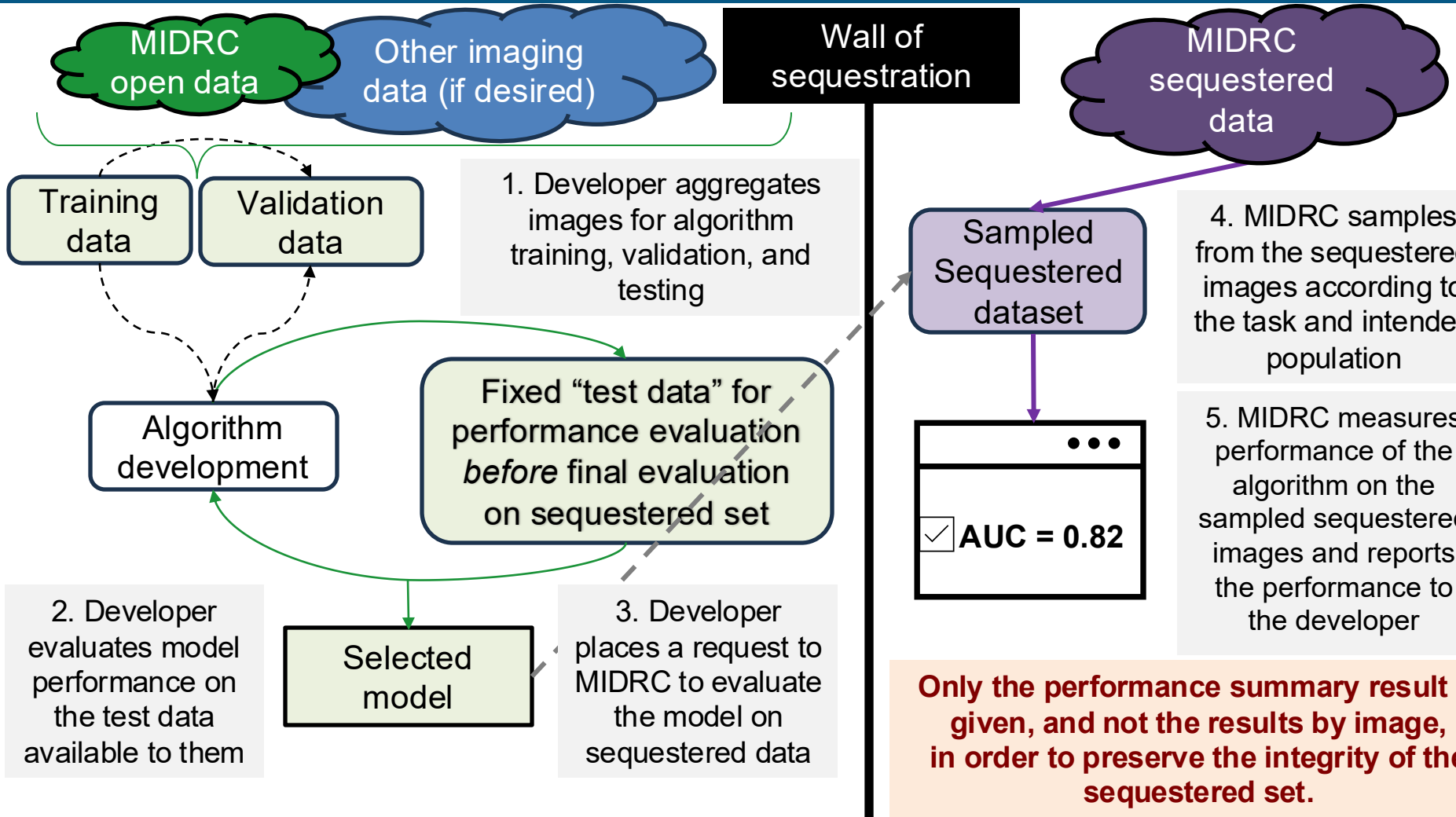
Baughan et al. "Sequestration of imaging studies in MIDRC: stratified sampling to balance demographic characteristics of patients in a multi-institutional data commons." *JMI* (2023)

Public facing DATA

- AI/ML ready
- FAIR
- Trustworthy
- Representative
- Searchable

Sequestered DATA

- **Not publicly available**
- **Real-world testing of algorithm performance**
- **Translation / support of regulatory submission**


MIDRC open data
Other imaging data (if desired)
Wall of sequestration
MIDRC sequestered data
Training data
Validation data
1. Developer aggregates images for algorithm training, validation, and testing
Sampled Sequestered dataset
4. MIDRC samples from the sequestered images according to the task and intended population
Algorithm development
Fixed "test data" for performance evaluation *before* final evaluation on sequestered set
AUC = 0.82
5. MIDRC measures performance of the algorithm on the sampled sequestered images and reports the performance to the developer
2. Developer evaluates model performance on the test data available to them
Selected model
3. Developer places a request to MIDRC to evaluate the model on sequestered data
Only the performance summary result is given, and not the results by image, in order to preserve the integrity of the sequestered set.

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Motivation

		Type of issue	
		Good faith	Fraudulent
Focus of this study	Sequestered	**A single evaluation of an algorithm could over or underestimate performance**	Developers try to learn about the sequestered test set through repeated evaluation
Data type	Open	Developers inadvertently have data leakage between training, validation, and their test sets	Developers may game the statistics used to define what is a ”better” result

Possible concerns around performance evaluation

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Motivation

Type of issue

Data type	Ingenuous	Disingenuous
Sequestered	**A single evaluation of an algorithm could over or underestimate performance**	Developers try to learn about the sequestered test set through repeated evaluation
Open	Developers inadvertently have data leakage between training, validation, and their test sets	Developers may game the statistics used to define what is a ”better” result

Possible concerns around performance evaluation

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Objectives of the study

The purpose of our study was to develop a method to measure medical imaging data use in test subsets and use it to

1. measure image reuse in test subsets

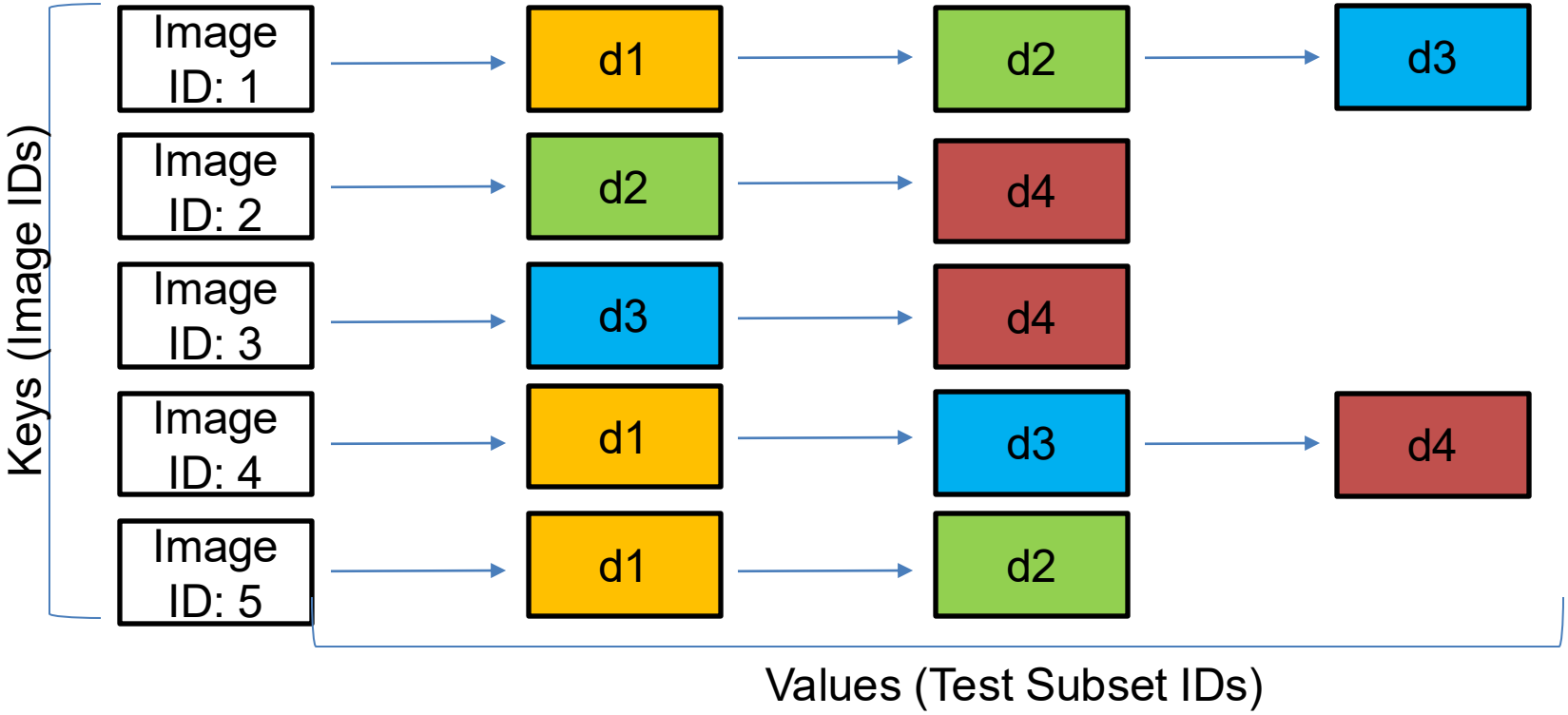
2. evaluate the impact of degree of image reuse on over- or under-estimation of performance.

3. compare these findings with expected AUC standard error in context of image reuse

Methods

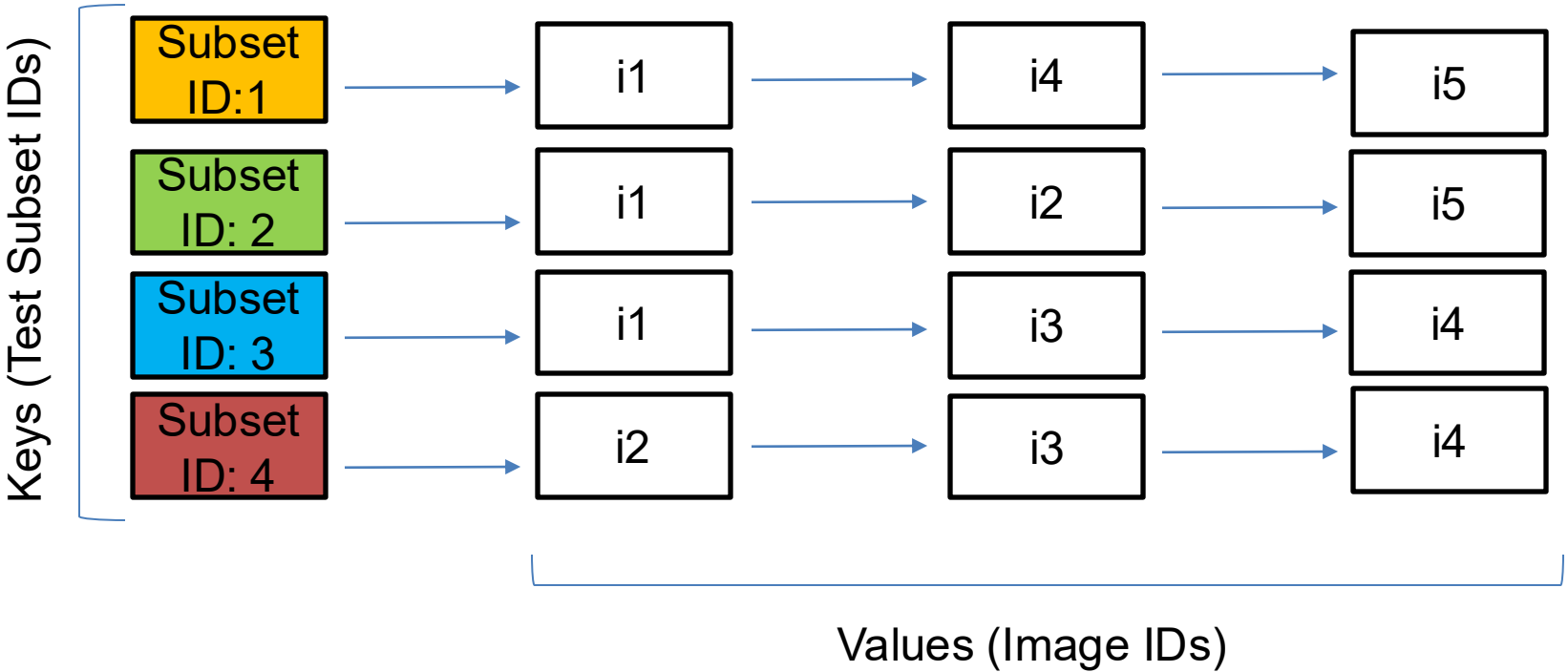
Methods: Hash Table Implementation

Conceptual overview of a hash table for the example of mapping **five** images to **four** test subsets of size **three**



Methods: Inversion of Hash Table

Inversion of the hash table, which now maps from subset IDs to image IDs



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Methods: Load Factor Analysis

- The load factor is a ratio of test subsets to images and serves as a metric that describes the average number of images drawn per test subset.

- The load factor can serve as a metric that summarizes the state of the distribution process.

$$\text{load factor } (\alpha) = \frac{(\text{number of test subsets} \cdot \text{test subset size})}{\text{number of images}}$$

High load factor: An image is likely used in **many** subsets

Low load factor: An image is likely used in **few** subsets

Methods: Analytical Relationship between Performance Variation and Load Factor

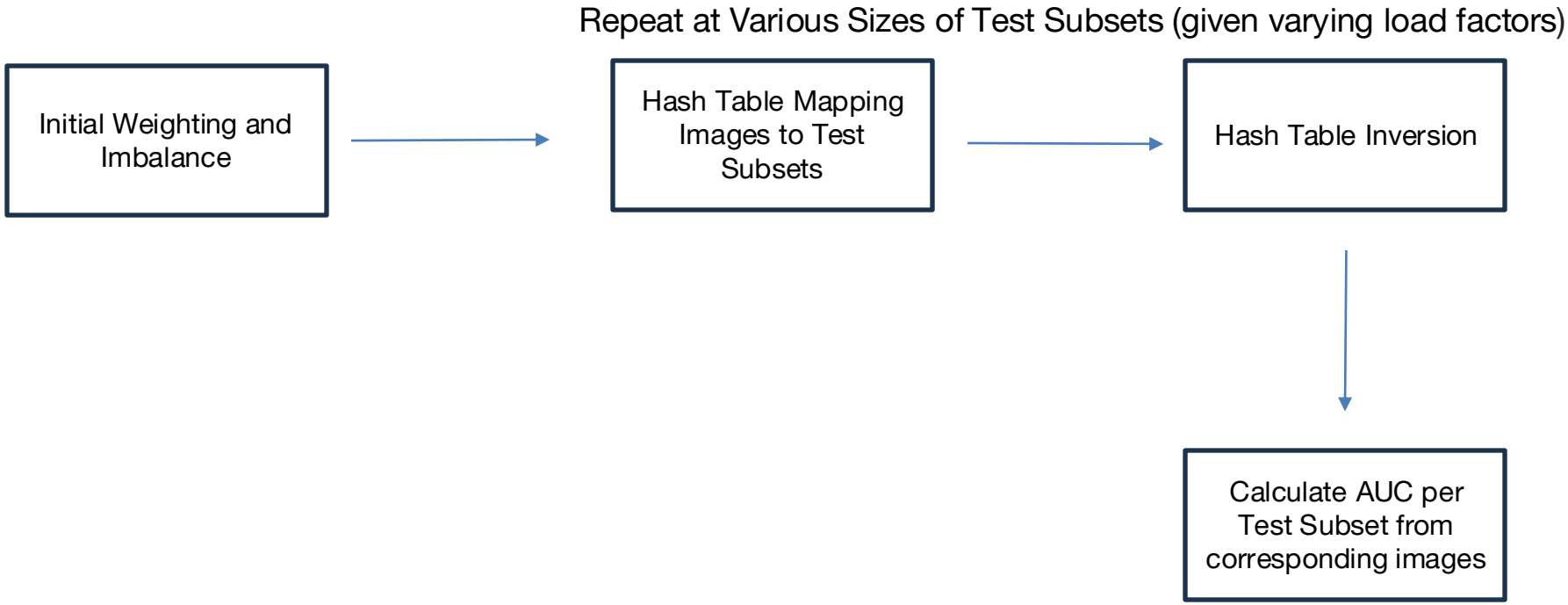
Load factor (α) is related to the standard error of the area under the curve (AUC) (θ):

$$SE(\theta,\alpha) = \sqrt{\frac{\hat{\theta}\left(1-\hat{\theta}\right) + \left(\frac{\alpha\cdot\phi}{\beta}\cdot i_A\right)\left(\frac{\hat{\theta}}{2-\hat{\theta}} - \hat{\theta}^2\right) + \left(\left(\frac{\alpha\cdot\phi}{\beta} - \frac{\alpha\cdot\phi}{\beta} - i_A\right) - 1\right)\left(\frac{2\hat{\theta}^2}{1+\hat{\theta}} - \hat{\theta}^2\right)}{\left(\frac{\alpha\cdot\phi}{\beta}\cdot i_A\right)\left(\frac{\alpha\cdot\phi}{\beta} - \frac{\alpha\cdot\phi}{\beta}\cdot i_A\right)}}$$

where $\hat{\theta}$ is the AUC of the entire dataset, α is the load factor, β is the number of samples, ϕ is the number of images, and i_A is the set prevalence of the minority class.

- We used this analytical expression to develop a theoretical estimate of over- or under-estimation of classification based upon load factor and prevalence.

Overview of Methods



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Clinical Data Study

- A dataset of 1047 chest radiograph (CXR) images from COVID-19+ patients had been acquired between March and September of 2020.
- Each image was associated with the corresponding status of the patient as admitted to the intensive care unit (ICU) or not within 24 hours of imaging. 12% of patients had been admitted to the ICU.
- A previously-trained deep learning/artificial intelligence model using DenseNet121 architecture had been used used to predict patients' need for intensive care within 24 hours of imaging.
- The area under the receiver operating characteristic curve (AUC) in the task of predicting patient ICU admission within 24 hours was AUC = 0.81.

Methods: Evaluation

(1) AUC versus load factor at (a) prevalence of clinical data and (b) substantially different prevalence

- Goal: to determine a potential relationship between load factor and AUC

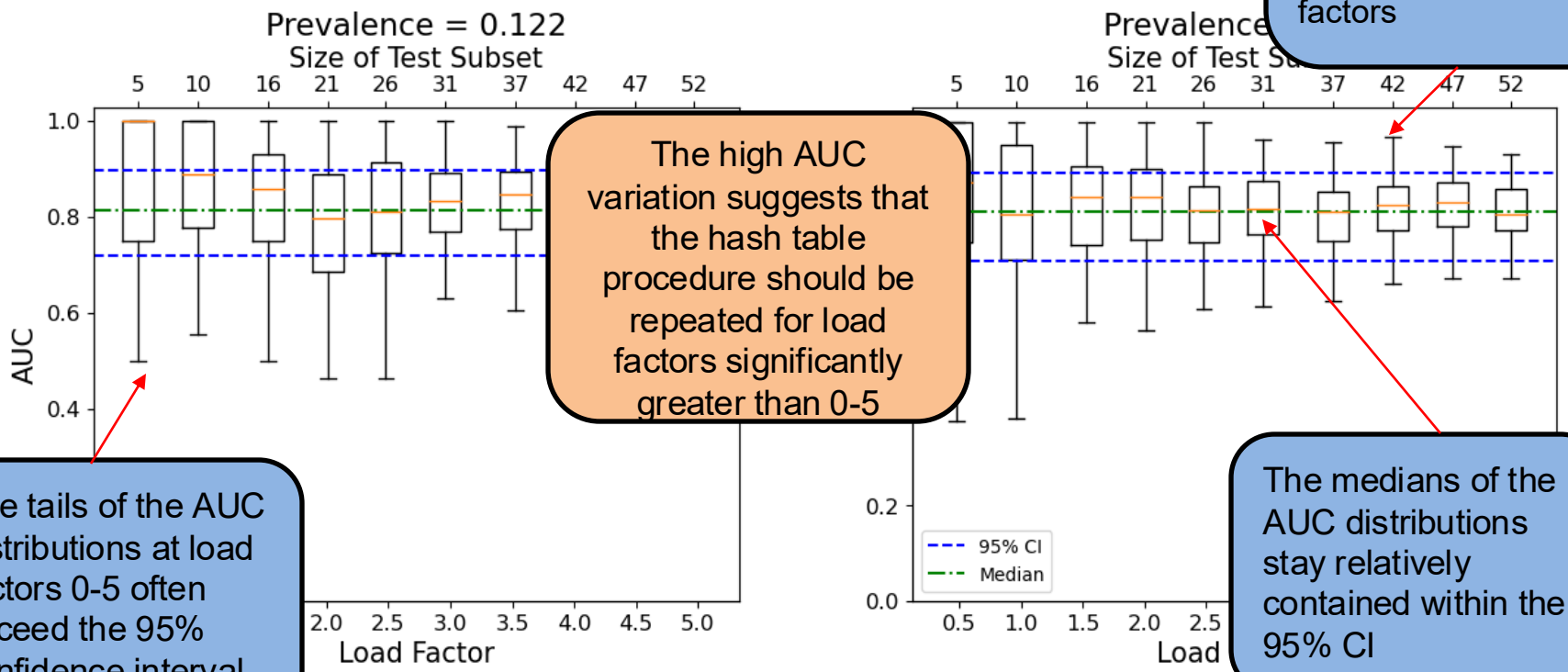
(2) Standard error (analytical prediction) and interquartile range (from clinical data) of AUC versus load factor

- Goal: to assess whether load factor can be used to forecast the potential for bias

Results

Results: (1) AUC Performance at Load Factors 0-5

Subsets constructed with a minority prevalence of 0.7 rather than 0.122 tend to exhibit less variation at this range of load factors



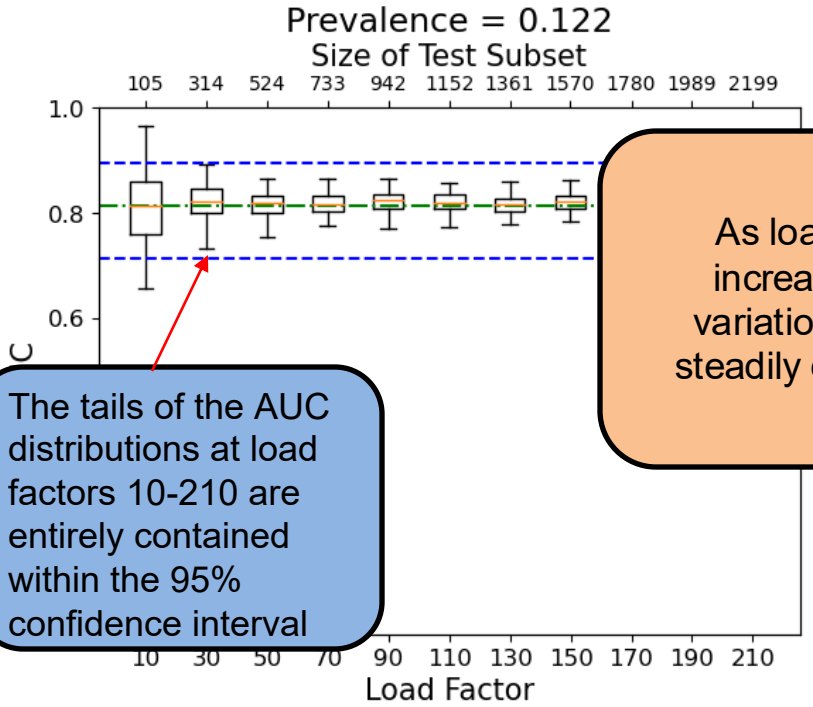
The high AUC variation suggests that the hash table procedure should be repeated for load factors significantly greater than 0-5

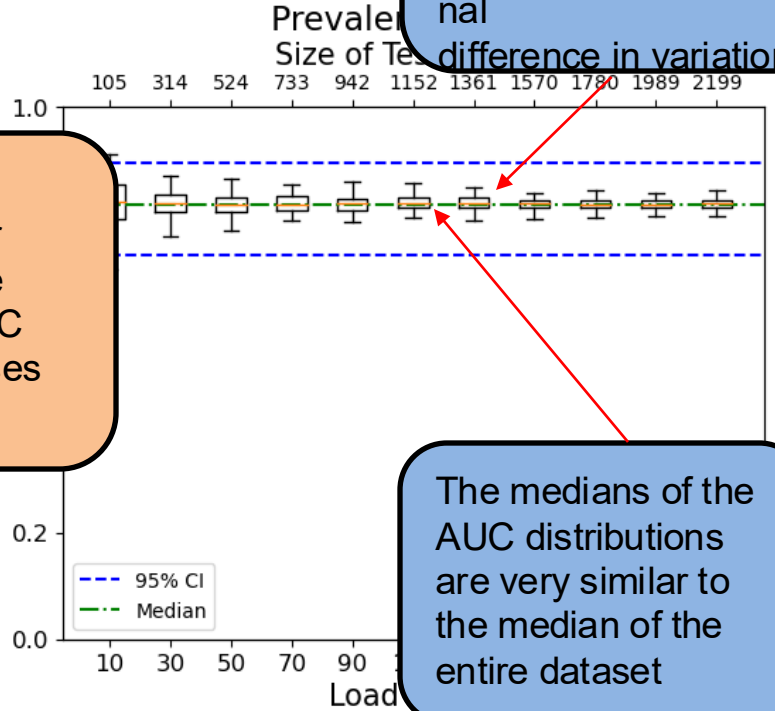
The tails of the AUC distributions at load factors 0-5 often exceed the 95% confidence interval

The medians of the AUC distributions stay relatively contained within the 95% CI

Results: (2) AUC Performance at Load Factors 10-210

Subsets constructed with a minority prevalence of 0.7 rather than 0.122 have marginal difference in variation





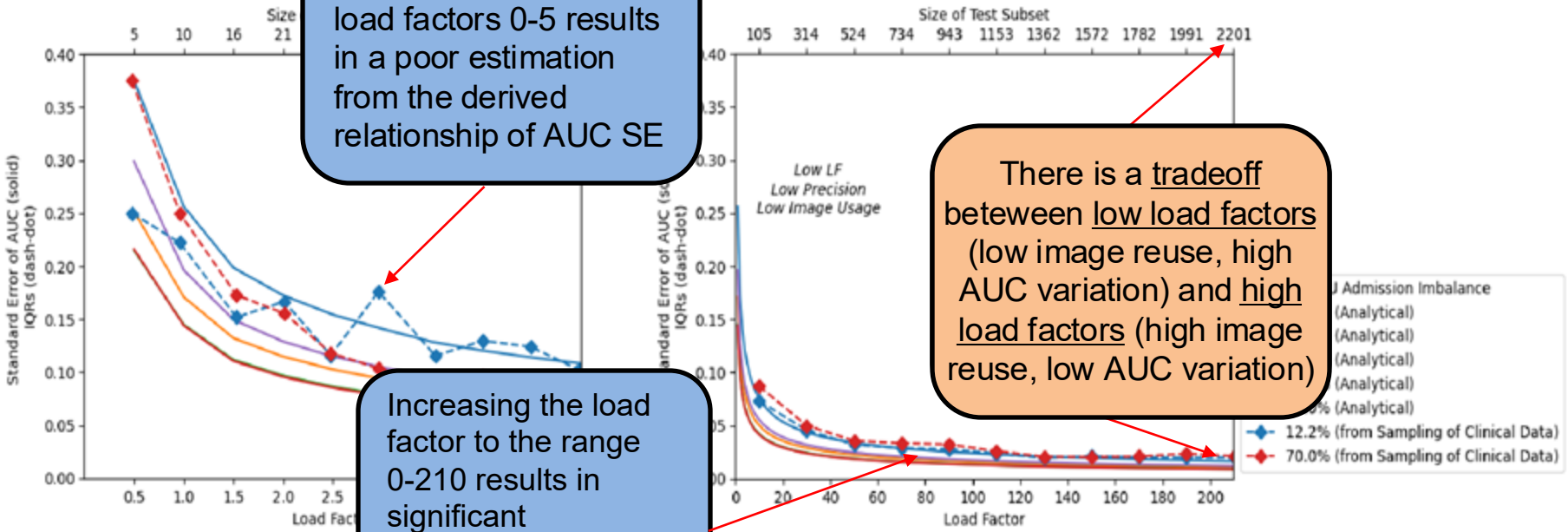
As load factor increases, the variation of AUC steadily decreases

The tails of the AUC distributions at load factors 10-210 are entirely contained within the 95% confidence interval

The medians of the AUC distributions are very similar to the median of the entire dataset

Results: (3) Estimated AUC Standard Error and Empirical IQRs with respect to Load Factor

The high variation of AUC IQR at load factors 0-5 results in a poor estimation from the derived relationship of AUC SE



There is a tradeoff beteween low load factors (low image reuse, high AUC variation) and high load factors (high image reuse, low AUC variation)

Increasing the load factor to the range 0-210 results in significant improvement in estimation

Conclusion

Conclusion

From our results, we conclude that the load factor derived from a hash-table implementation is

1. A concise yet powerful metric for evaluating the distribution of images that takes advantage of the natural pairing between images and test subsets
2. A metric that captures the tradeoff between variation and image reuse

These results point to the need of further investigations of realizing the potential of load factor for evaluating the distribution process of images/datasets.

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Share your imaging data with MIDRC to satisfy the 2023 NIH DMS policy for

NIH

Scientific Data Sharing

Contact the MIDRC team **midrc-support-request@lists.uchicago.edu**
to learn more about how we can support you in data sharing visit **midrc.org**

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Thank you

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